SENTHIL PITCHAPPAN V

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PROFESSIONAL EXPERIENCE

ZOHO CORPORATION Chennai, India

Member Technical Staff, Backend Engineer, CPaaS and Distributed Messaging Platform Mar 2021 - Present

- Designed and delivered a unified, omnichannel CPaaS framework integrating multiple channels, including WhatsApp Business Platform, into one RESTful API surface, standardising and speeding up new-channel onboarding
- Architected and built the backend of the **Arattai Business Platform** from the ground up (data modelling, microservice architecture, message workflows and credit management system), scaling to millions of messages in production
- Engineered WebRTC-based infrastructure for **WhatsApp Business Calling**, owning system design, data model, APIs, scalability, testing and production rollout, and integrated real-time calling into the messaging platform
- Developed and owned an external bot microservice in Java on Kafka and Redis, processing high-throughput message workloads at low p99 latency with high availability for several internal Zoho products
- Built event-driven, multi-threaded Kafka pipelines with consumer-group scaling, partition-aware routing and back-pressure handling, delivering fault-tolerant, at-least-once processing across tenants
- Authored an end-to-end acceptance test suite (JUnit, WireMock) mocking carrier and partner APIs, reducing production regressions; led design reviews, ran services on Kubernetes and mentored junior engineers

MARVEL TECHNOLOGIES PVT. LTD. Madurai, India

Software Developer Intern Dec 2018 - Jun 2019

- Built an IoT freezer monitoring and control system (Raspberry Pi, Arduino) with real-time alerts and ERP front-end modules
- Received the Star Performer Award for project delivery and cross-team collaboration

PROJECTS

Nudgeway (Live Demo)

Open-source multi-tenant WhatsApp Business messaging platform (Go, MySQL, Redis, Kafka) 2026

- Built a provider-agnostic Go modular monolith: WhatsApp messaging with delivery-status tracking, WebSocket live inbox, template CRUD with Meta sync, media in HBase, and voice calling via in-browser WebRTC; 38 docs-site visits in its first five weeks

WhatsApp Business Agent Studio (Live Demo)

Visual builder for WhatsApp AI agents (Next.js, TypeScript, Meta Graph API) 2026

- Wrapped Meta's Business Agent Platform API in one workspace: WABA discovery, agent onboarding, skills, knowledge, tool connectors, evals, chat-testing and webhooks; 281 visits across 12+ pages in the first three months after launch

Newt

Personal knowledge base with local AI chat (Node.js, Fastify, SQLite FTS5, Claude Agent SDK) 2026

- Created a Markdown vault with SQLite FTS5 search, Claude Code slash commands for capture and cited recall, a localhost SPA (dashboard, scoped docs chat, session workbench, work journal) and multi-engine chat (Claude, Ollama, LM Studio) over 325 docs

LearningRAG

RAG pipeline built from scratch, no frameworks (Python, Ollama, FastAPI, Streamlit) 2026

- Implemented embeddings, chunking strategies, vector search, BM25 from scratch and hybrid retrieval, evaluated on a 23-query labelled set; shipped ingest and Q&A CLIs plus a FastAPI server with streaming, model switching and a UMAP embedding map

InkCounter, MarkViewer, JSONForge and Developer Tools

TypeScript, React, Node.js, PostgreSQL, CodeMirror 6, Monaco, React Flow 2026

- Shipped self-hosted web analytics (Docker Compose), an AI Markdown IDE (OpenAI, Gemini, Ollama), a 20 MB-capable JSON formatter with Web Worker parsing and graph view, an expense tracker and a trip planner

College and Early Projects

Python, Flask, Django, OpenCV, PHP, Arduino, ESP32, Raspberry Pi, Jupyter 2019 - 2021

- Automatic Remote Monitoring System (final-year project, Flask); Smart Garbage Management System (IoT); number-plate detection (OpenCV); IoT, storm-prediction and device-tracker dashboards (Flask); Django REST apps; Arduino and ESP32 projects

TECHNICAL SKILLS

- **Languages:** Java (primary), Python, SQL, Go and TypeScript (projects)
- **Backend & Distributed Systems:** Microservices, event-driven architecture, Apache Kafka, multi-threading and concurrency, high-throughput and low-latency design, scalability, fault tolerance, multi-tenancy, idempotency, rate limiting, caching
- **Messaging, Data & Infrastructure:** CPaaS, WhatsApp Business Platform (Cloud API), webhooks, WebRTC; MySQL, PostgreSQL, HBase, ClickHouse, Redis; Kubernetes, Docker, Apache Tomcat, Linux, Git, RESTful API design, observability
- **Testing, Practices & AI Tooling:** JUnit, WireMock, Cucumber, contract testing, system design, data modelling, performance tuning, design patterns, code reviews; Claude, Cursor, GitHub Copilot, LLMs, SLMs, open-source models, MCP, AI agents, RAG

EDUCATION

VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY Madurai, India

Bachelor of Engineering, Electronics and Communication Engineering Mar 2017 - May 2021

- Smart India Hackathon (Software Edition), Government of India: national finalist, Panipat, 2019
- Published “Smart Robot for Defence and Surveillance using IoT” in the Journal of Emerging Technologies and Innovative Research